Year 11 Higher

REVISION PACK

Progress Check 4

Thursday 26 November 2026

Name: _____

Class: _____

Units in this pack

- Graphs 6
- Handling Data 5

Graphs 6 · Graphs to solve equations (Lessons 8.1, 8.2)

Worked example: solving with a quadratic graph

The graph of $y = x^2 - 2x - 2$ is drawn on a grid.

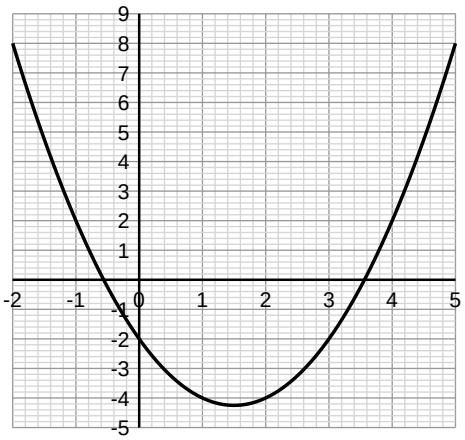
(a) Use the graph to find estimates for the solutions of $x^2 - 2x - 2 = 0$

(b) By drawing a suitable straight line, find estimates for the solutions of $x^2 - 3x - 1 = 0$

(a)	**Step 1**	Read where the curve crosses the x-axis	
	Step 2	Answer	$x = -0.7$ or $x = 2.7$
(b)	**Step 1**	Make the left side the curve	$x^2 - 2x - 2 = x - 1$
	Step 2	Draw the line	$y = x - 1$
	Step 3	Read where they cross	
	Step 4	Answer	$x = -0.3$ or $x = 3.3$

Practice: solving with a quadratic graph

1.1

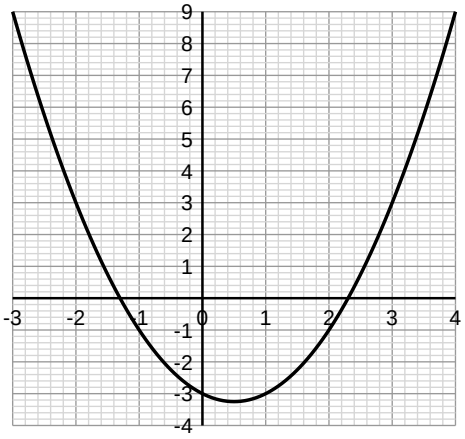


The graph of $y = x^2 - 3x - 2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 - 3x - 2 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 2x - 4 = 0$

1.2



The graph of $y = x^2 - x - 3$ is drawn on the grid.

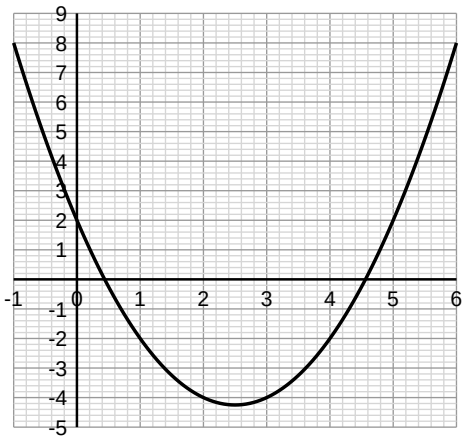
(a) Use the graph to find estimates for the solutions of $x^2 - x - 3 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 2x - 1 = 0$

Graphs 6 · Graphs to solve equations (Lessons 8.1, 8.2)

Practice: solving with a quadratic graph (continued)

1.3

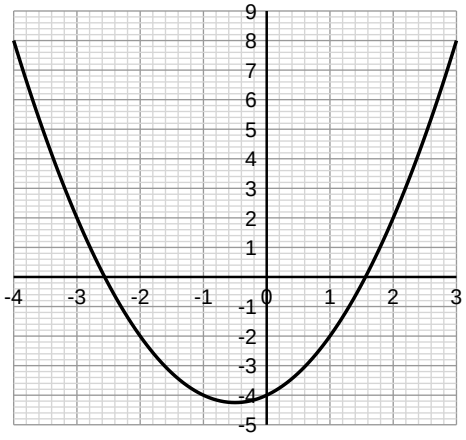


The graph of $y = x^2 - 5x + 2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 - 5x + 2 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 4x - 1 = 0$

1.4



The graph of $y = x^2 + x - 4$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^2 + x - 4 = 0$

(b) By drawing a suitable straight line on the grid, find estimates for the solutions of $x^2 - 2 = 0$

Graphs 6 · Graphs to solve equations (Lessons 8.1, 8.2)

Worked example: solving with a cubic graph

The graph of $y = x^3 - 3x + 2$ is drawn on a grid.

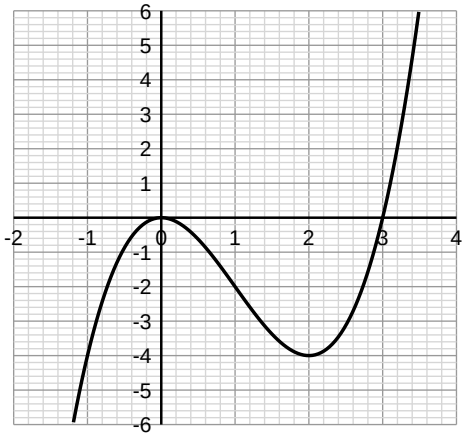
(a) Find estimates for the solutions of $x^3 - 3x + 2 = 1$

(b) The equation $x^3 - 3x + 2 = k$ has three solutions. Write down the range of values of k.

(a)	Step 1	Draw the line	$y = 1$
	Step 2	Read where they cross	
	Step 3	Answer	$x = -1.9,\ 0.3,\ 1.5$
(b)	Step 1	Read the turning points	$y = 4$ and $y = 0$
	Step 2	Answer	$0 < k < 4$

Practice: solving with a cubic graph

1.5

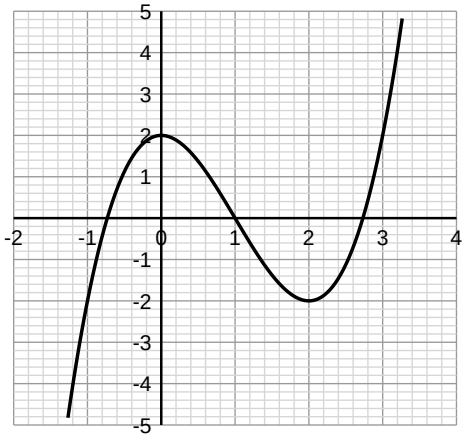


The graph of $y = x^3 - 3x^2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^3 - 3x^2 = -2$

(b) The equation $x^3 - 3x^2 = k$ has three solutions. Write down the range of possible values of k.

1.6



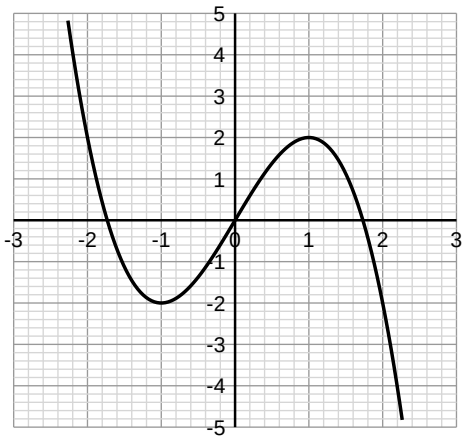
The graph of $y = x^3 - 3x^2 + 2$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^3 - 3x^2 + 2 = 1$

(b) The equation $x^3 - 3x^2 + 2 = k$ has three solutions. Write down the range of possible values of k.

Practice: solving with a cubic graph (continued)

1.7

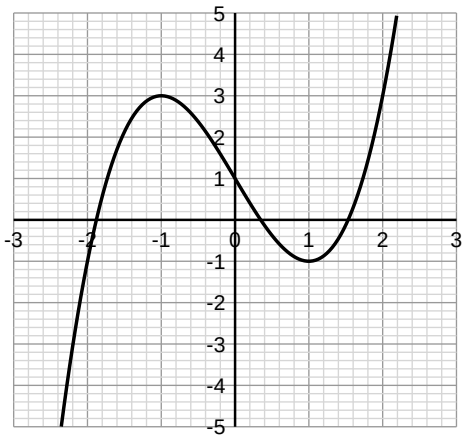


The graph of $y = 3x - x^3$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $3x - x^3 = -1$

(b) The equation $3x - x^3 = k$ has three solutions. Write down the range of possible values of k.

1.8



The graph of $y = x^3 - 3x + 1$ is drawn on the grid.

(a) Use the graph to find estimates for the solutions of $x^3 - 3x + 1 = 1$

(b) The equation $x^3 - 3x + 1 = k$ has three solutions. Write down the range of possible values of k.

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Worked example: a line and a circle

Solve the simultaneous equations $y = x + 7$ and $x^2 + y^2 = 25$

Step 1	Substitute	$x^2 + (x+7)^2 = 25$
Step 2	Expand	$x^2 + x^2 + 14x + 49 = 25$
Step 3	Simplify	$2x^2 + 14x + 24 = 0$
Step 4	Divide by 2	$x^2 + 7x + 12 = 0$
Step 5	Factorise	$(x+3)(x+4) = 0$
Step 6	Find y	$y = -3 + 7 = 4, \quad y = -4 + 7 = 3$
Step 7	Answer	$x = -3, y = 4$ or $x = -4, y = 3$

Practice: a line and a circle

2.1 Solve the simultaneous equations

$y = x - 1$

$x^2 + y^2 = 13$

2.2 Solve the simultaneous equations

$y = 2x - 5$

$x^2 + y^2 = 10$

2.3 Solve the simultaneous equations

$y = x + 3$

$x^2 + y^2 = 17$

2.4 Solve the simultaneous equations

$y = -x + 5$

$x^2 + y^2 = 13$

2.5 Solve the simultaneous equations

$y = 3x - 10$

$x^2 + y^2 = 20$

2.6 Solve the simultaneous equations

$y = x - 2$

$x^2 + y^2 = 10$

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

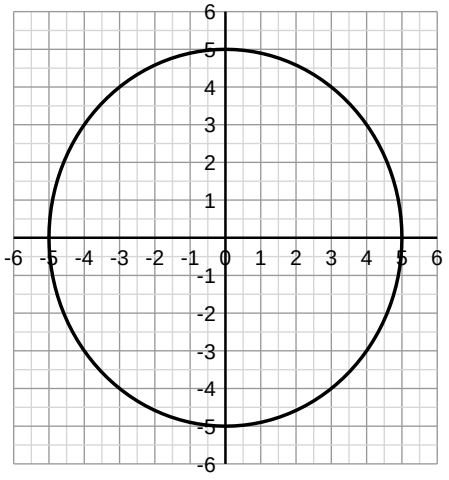
Worked example: a line and a circle on a graph

The graph of $x^2 + y^2 = 25$ is drawn on a grid. By drawing a suitable straight line, solve $x^2 + y^2 = 25$ and $y = x - 7$

Step 1	Draw the line	$y = x - 7$
Step 2	Read where they cross	$(3, -4)$ and $(4, -3)$
Step 3	Answer	$x = 3, y = -4$ or $x = 4, y = -3$

Practice: a line and a circle on a graph

2.7



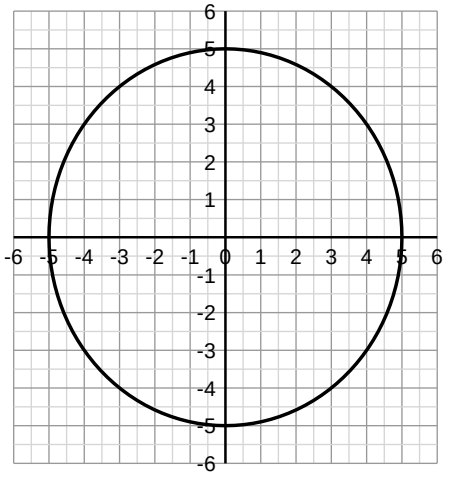
The graph of $x^2 + y^2 = 25$ is drawn on the grid.

By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 25$

$y = x - 1$

2.8



The graph of $x^2 + y^2 = 25$ is drawn on the grid.

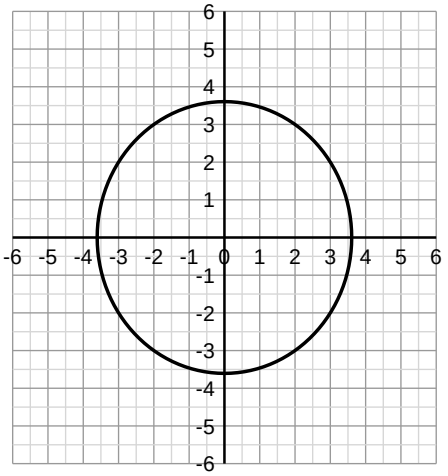
By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 25$

$y = -x + 7$

Practice: a line and a circle on a graph (continued)

2.9



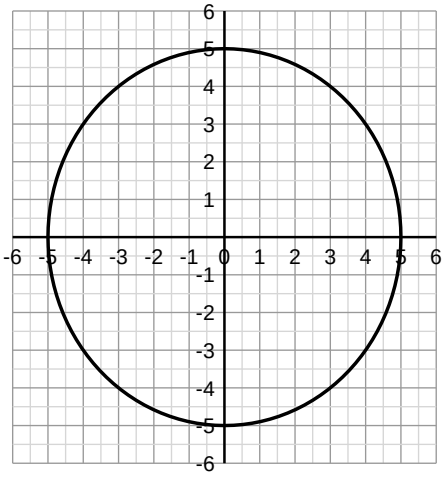
The graph of $x^2 + y^2 = 13$ is drawn on the grid.

By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 13$

$y = x + 1$

2.10



The graph of $x^2 + y^2 = 25$ is drawn on the grid.

By drawing a suitable straight line on the grid, find the solutions of the simultaneous equations

$x^2 + y^2 = 25$

$y = -x + 1$

Handling Data 5 · Averages from frequency tables (Lesson 9.1)

Worked example: grouped data

The table shows the times, in minutes, that 20 people waited.

Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$
Frequency	4	10	6

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

(a)	**Step 1**	Highest frequency	10
	Step 2	Answer	$10 < t \leq 20$
(b)	**Step 1**	Find the middle	$(20 + 1) \div 2 = 10.5\text{th}$
	Step 2	Add the frequencies	$4,\ 14$
	Step 3	Answer	$10 < t \leq 20$
(c)	**Step 1**	Multiply midpoints and add	$5 \times 4 + 15 \times 10 + 25 \times 6 = 320$
	Step 2	Divide by the total	$320 \div 20 = 16$
	Step 3	Answer	$16\ \text{minutes}$

Practice: grouped data

3.1 The table shows the times, in minutes, that 40 people took to travel to school.

Time (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$
Frequency	4	9	15	8	4

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

3.2 The table shows the masses, in grams, of 50 apples.

Mass (m grams)	$0 < m \leq 20$	$20 < m \leq 40$	$40 < m \leq 60$	$60 < m \leq 80$	$80 < m \leq 100$
Frequency	6	14	20	7	3

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

3.3 The table shows the waiting times, in minutes, of 40 patients.

Time (t minutes)	$0 < t \leq 5$	$5 < t \leq 10$	$10 < t \leq 15$	$15 < t \leq 20$	$20 < t \leq 25$
Frequency	3	11	12	10	4

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

Practice: grouped data (continued)

3.4 The table shows the lengths, in minutes, of 40 phone calls.

Length (t minutes)	$0 < t \leq 10$	$10 < t \leq 20$	$20 < t \leq 30$	$30 < t \leq 40$	$40 < t \leq 50$
Frequency	2	8	14	11	5

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

3.5 The table shows the times, in seconds, that 40 students took to answer a puzzle.

Time (t seconds)	$0 < t \leq 4$	$4 < t \leq 8$	$8 < t \leq 12$	$12 < t \leq 16$	$16 < t \leq 20$
Frequency	7	13	9	6	5

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

3.6 The table shows the ages, in years, of 50 people at a party.

Age (a years)	$0 < a \leq 10$	$10 < a \leq 20$	$20 < a \leq 30$	$30 < a \leq 40$	$40 < a \leq 50$
Frequency	5	10	15	12	8

(a) Write down the modal class.

(b) Write down the class interval that contains the median.

(c) Work out an estimate for the mean.

Handling Data 5 · Averages from frequency tables (Lesson 9.1)

Worked example: ungrouped data

The table shows the number of goals scored in 10 matches.

Number of goals	0	1	2
Frequency	3	5	2

(a) Work out the mean. (b) Find the median.

(a)	Step 1	Multiply and add	$0 \times 3 + 1 \times 5 + 2 \times 2 = 9$
	Step 2	Divide by the total	$9 \div 10 = 0.9$
	Step 3	Answer	0.9
(b)	Step 1	Find the middle	$(10 + 1) \div 2 = 5.5\text{th}$
	Step 2	Add the frequencies	$3,\ 8$
	Step 3	Answer	1

Practice: ungrouped data

3.7 The table shows the number of goals scored by a team in 20 matches.

Number of goals	0	1	2	3	4
Frequency	3	6	5	4	2

(a) Work out the mean. (b) Find the median.

3.8 The table shows the number of people living in each of 20 houses.

Number of people	1	2	3	4	5
Frequency	4	7	5	3	1

(a) Work out the mean. (b) Find the median.

3.9 The table shows the number of pets owned by each of 20 students.

Number of pets	0	1	2	3
Frequency	5	8	4	3

(a) Work out the mean. (b) Find the median.

3.10 The table shows the number of books read by each of 20 students in a month.

Number of books	0	1	2	3	4
Frequency	2	5	8	3	2

(a) Work out the mean. (b) Find the median.

3.11 The table shows the number of cars owned by each of 20 families.

Number of cars	1	2	3	4
Frequency	6	9	3	2

(a) Work out the mean. (b) Find the median.

3.12 The table shows the number of brothers and sisters of each of 20 students.

Number	0	1	2	3	4	5
Frequency	1	4	6	5	3	1

(a) Work out the mean. (b) Find the median.

Graphs 6 · Non-linear simultaneous equations (Lesson 8.3)

Worked example: a line and a quadratic

Solve the simultaneous equations $y = 2x + 1$ and $y = x^2 - 2$

Step 1	Make them equal	$x^2 - 2 = 2x + 1$
Step 2	Make one side 0	$x^2 - 2x - 3 = 0$
Step 3	Factorise	$(x - 3)(x + 1) = 0$
Step 4	Solve	$x = 3$ or $x = -1$
Step 5	Find y	$y = 2 \times 3 + 1 = 7, \quad y = 2 \times (-1) + 1 = -1$
Step 6	Answer	$x = 3, y = 7$ or $x = -1, y = -1$

Practice: a line and a quadratic

4.1 Solve the simultaneous equations

$y = x + 5$

$y = x^2 - 1$

4.2 Solve the simultaneous equations

$y = 2x + 3$

$y = x^2$

4.3 Solve the simultaneous equations

$y = 3x - 2$

$y = x^2 - 2x + 4$

4.4 Solve the simultaneous equations

$y = -x + 4$

$y = x^2 - 2$

4.5 Solve the simultaneous equations

$y = 2x - 1$

$y = x^2 + x - 3$

4.6 Solve the simultaneous equations

$y = 4x - 3$

$y = x^2$